UiA Universitetet i Agder

# MAS508-G
# Lab 2 – System modeling and simulation solution



# Example 1: System differential equations

Kirchhoff Voltage Law (KLV) for the circuit:

$$u - Ri - L\frac{di}{dt} - e = 0$$

Induced voltage is proportional to the rotational speed:

$$e = \Psi\omega$$

Moment balance for the rotational-mechanical system:

$$J\frac{dw}{dt} + Bw = M - M_L$$

Torque $M$ developed by the motor is proportional to the armature current

$$M = \Psi i$$

Assuming electrical motor without load, i.e. $M_L = 0$

$$J\frac{dw}{dt} + Bw = \Psi i$$

Model Equations

$$L\frac{di}{dt} + Ri = u - \Psi\omega$$

$$J\frac{dw}{dt} + Bw = \Psi i$$

# Example 1: Block diagram

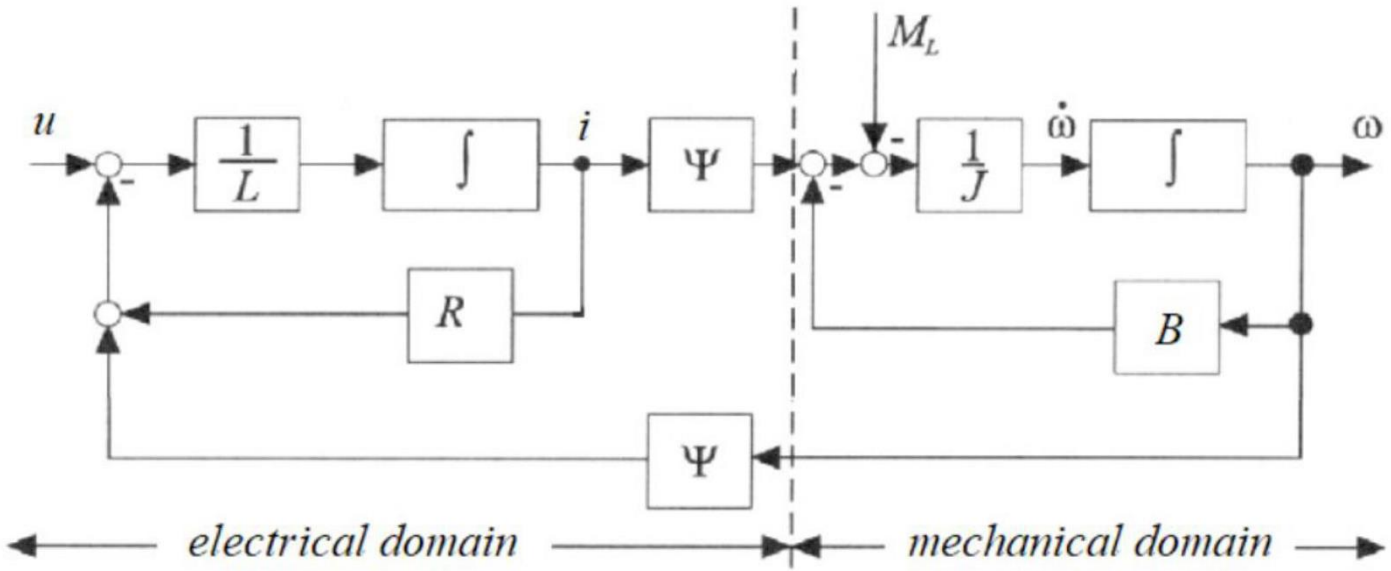
**Model equations:**

$$L\frac{di}{dt} + Ri = u - \Psi\omega$$

$$J\frac{dw}{dt} + Bw = \Psi i - M_L$$

Laplace Transformation (without disturbance)

$$L \cdot s \cdot I(s) + R \cdot I(s) = U(s) - \Psi\Omega(s)$$
$$J \cdot s \cdot \Omega(s) + B \cdot \Omega(s) = \Psi I(s)$$



# Example 1: Transfer function

$$\begin{cases} L \cdot s \cdot I(s) + R \cdot I(s) = U(s) - \Psi\Omega(s) \\ J \cdot s \cdot \Omega(s) + B \cdot \Omega(s) = \Psi I(s) \end{cases} \Leftrightarrow \begin{cases} (Ls+R)I(s) = U(s) - \Psi\Omega(s) \\ (Js+B)\Omega(s) = \Psi I(s) \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} U(s) = (Ls+R)I(s) - \Psi\Omega(s) \\ \Omega(s) = \frac{\Psi I(s)}{Js+B} \end{cases}$$

$$\Rightarrow \frac{\Omega(s)}{U(s)} = \frac{\frac{\Psi I(s)}{Js+B}}{(Ls+R)I(s) + \Psi\Omega(s)}$$

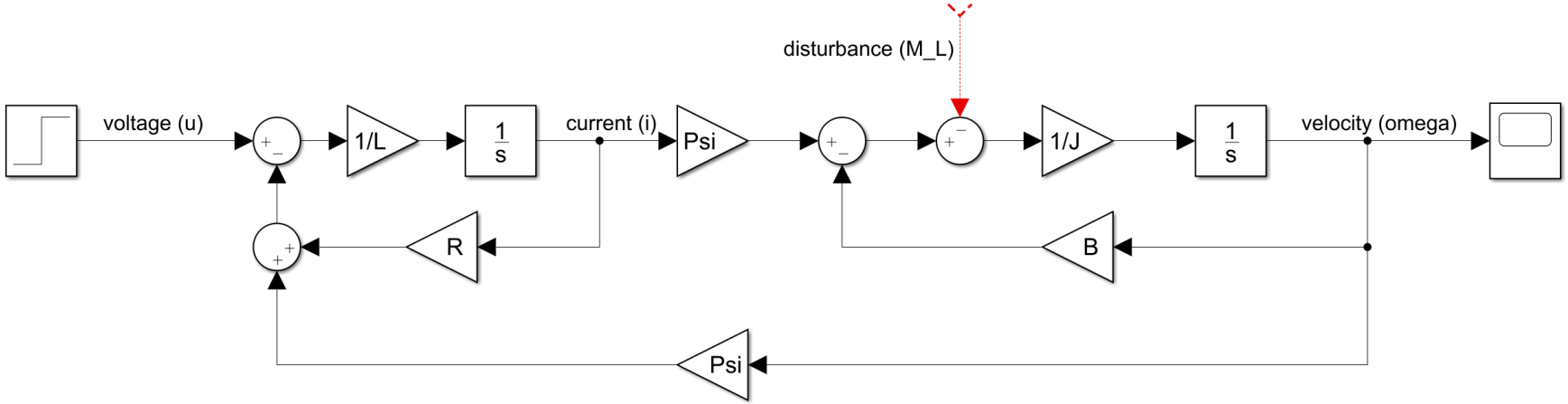
$$\Leftrightarrow \frac{\Omega(s)}{U(s)} = \frac{\Psi I(s)}{(Js+B)(Ls+R)I(s) + (Js+B)\Psi\Omega(s)}$$

$$\Leftrightarrow \frac{\Omega(s)}{U(s)} = \frac{\Psi I(s)}{(Js+B)(Ls+R)I(s) + (Js+B)\frac{\Psi^2 I(s)}{Js+B}}$$

$$\Leftrightarrow \frac{\Omega(s)}{U(s)} = \frac{\Psi}{(Js+B)(Ls+R) + \Psi^2}$$

$$\Leftrightarrow \frac{\Omega(s)}{U(s)} = \frac{\Psi}{JLs^2 + (JR+BL)s + BR + \Psi^2}$$

# Example 1: MATLAB/ Simulink




## MATLAB code (transfer function):

```
%parameters
R=1; L=0.0002; Psi=0.04; B=0.0001; J=0.00005;

%Transfer function
tf1=tf([Psi],[J*L (J*R+B*L) B*R+Psi^2]); %tf(num,den)

%Step response
step(tf1,0.5); %t=0.5s

%Bode diagram
bode(tf1);
```

step() - calculates the step response of a dynamic system (e.g. transfer function object)
step(sys,Tfinal)

bode() - Bode plot of frequency response, or magnitude and phase data
bode(sys1)

# Example 2: Block diagram

Model equations:

$$J_l\ddot{\theta}_l + B_l\dot{\theta}_l + k(\theta_l - \theta_m) = 0$$
$$J_m\ddot{\theta}_m + B_m\dot{\theta}_m - k(\theta_l - \theta_m) = u$$

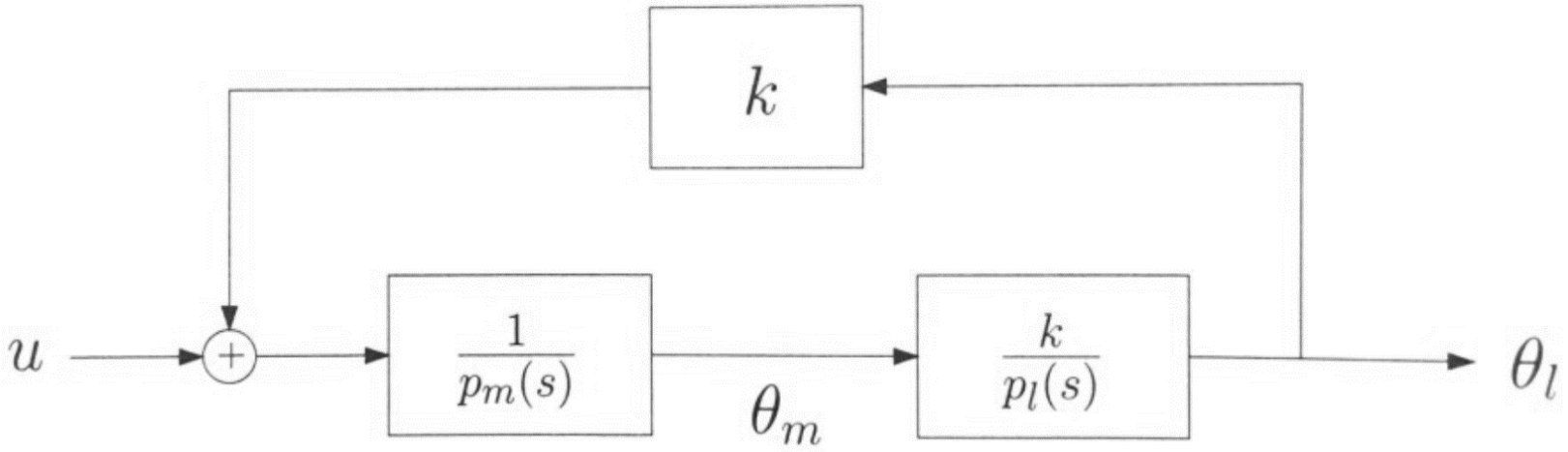
Laplace Transformation

$$p_l(s) = J_l s^2 + B_l s + k$$

$$p_m(s) = J_m s^2 + B_m s + k$$

$$p_l(s)\Theta_l(s) = k\Theta_m(s)$$

$$p_m(s)\Theta_m(s) = k\Theta_l(s) + U(s)$$



# Example 2: Transfer function

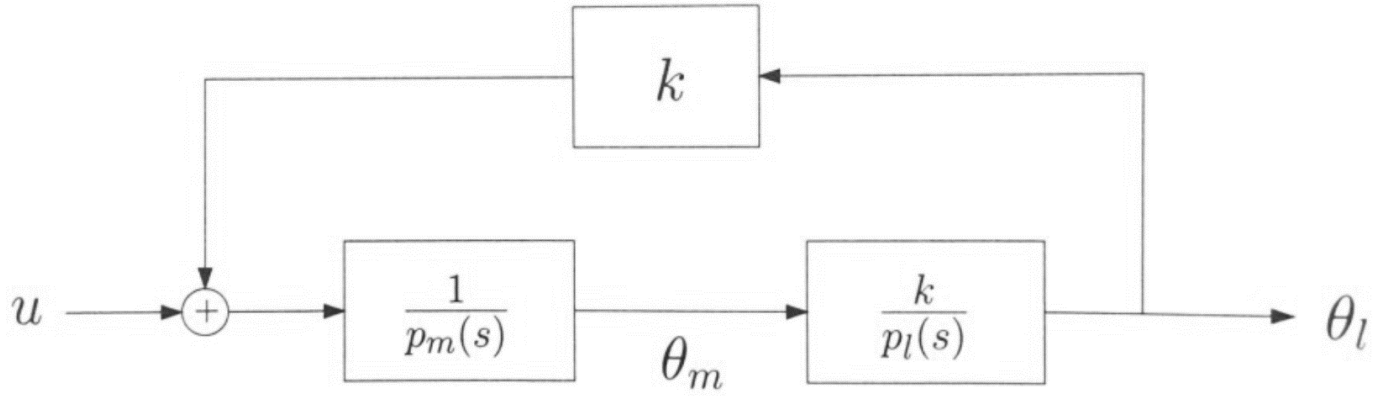
$$\begin{cases} p_l(s)\Theta_l(s) = k\Theta_m(s) \\ p_m(s)\Theta_m(s) = k\Theta_l(s) + U(s) \end{cases} \Leftrightarrow \begin{cases} \Theta_l(s) = \frac{k}{p_l(s)}\Theta_m(s) \\ U(s) = p_m(s) + \Theta_m(s) - k\Theta_l \end{cases}$$

$$\Rightarrow \frac{\Theta_l(s)}{U(s)} = \frac{\frac{k}{p_l(s)}\Theta_m(s)}{p_m(s)\Theta_m(s) - k\Theta_l(s)}$$

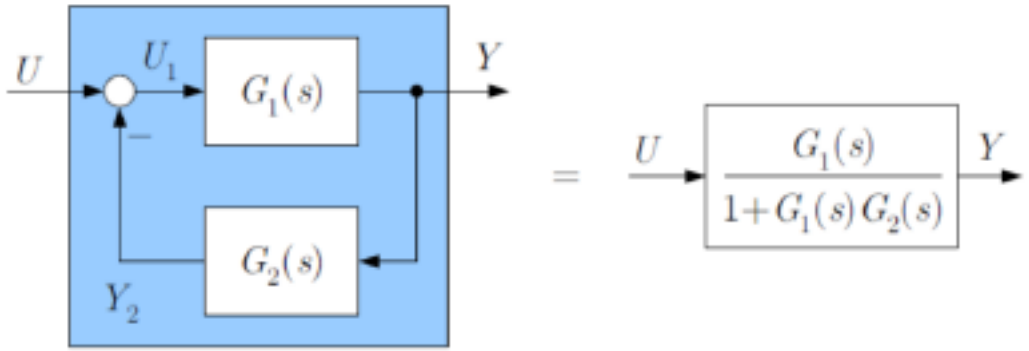
$$\frac{\Theta_l(s)}{U(s)} = \frac{\frac{k}{p_l(s)}}{p_m(s) - \frac{k^2}{p_l(s)}}$$

$$\Leftrightarrow \frac{\Theta_l(s)}{U(s)} = \frac{k}{p_m(s)p_l(s) - k^2}$$

# Example 2: Transfer function via block diagram (alternate method)




**Closing loop:**




$$\frac{\Theta_l(s)}{U(s)} = \frac{G_1(s)}{1 + G_1(s)G_2(s)}$$

$$G_1(s) = \frac{k}{p_m(s)p_l(s)}$$

$$G_2(s) = -k$$

$$\Leftrightarrow \frac{\Theta_l(s)}{U(s)} = \frac{\frac{k}{p_m(s)p_l(s)}}{1 - \frac{k}{p_m(s)p_l(s)}k}$$

$$\Leftrightarrow \frac{\Theta_l(s)}{U(s)} = \frac{k}{p_m(s)p_l(s) - k^2}$$

# Example 2: MATLAB

```
Jm=2; Bm=0.5; Jl=10; Bl=1; k=100;

%% Transfer function
tf1=tf([k],[(Jm*Jl) (Jm*Bl+Jl*Bm) (Jm*k+Jl*k+Bm*Bl) (Bm*k+Bl*k) 0]);

%alternative way to obtain transfer function (for verification)
pl=tf([Jl Bl k],[1]);
pm=tf([Jm Bm k],[1]);
G1=k/(pm*pl);
G2=-k;
tf2=feedback(G1,G2);
```

tf() - create a transfer function model object by specifying its coefficients
tf(numerator,denominator)

feedback() - returns a model object for the negative feedback interconnection of model objects sys1,sys2.
sys = feedback(sys1,sys2)

